CytoSet: Bethesda-Order-Aware Set Prediction for Cervical Cytology Cell Detection

Cheng He, Fufeng Chen, Fuhai Chen, Weiwei Guo, Chunxu Yang, Zhiwei Chen, and Qingqing Zhou

***Abstract*— Squamous cell detection in cervical ThinPrep Cytologic Test (TCT) images requires accurate localization and Bethesda category assignment. We represent the five squamous categories by an operational reporting order, yet standard detectors treat them as unordered labels and apply uniform classification objectives. This formulation ignores both crossed reporting boundaries and error direction: morphological overlap makes adjacent-grade confusion common, whereas distant or high-to-low errors may reduce the high-grade cellular evidence presented for downstream review. For set-prediction detectors, ordered supervision also depends on whether the matched queries provide reliable semantic representations and sufficient spatial coverage of candidate cells. We propose CytoSet, a Bethesda-order-aware adaptation of RT-DETR that conditions the query set before applying ordered learning. Dual-codebook semantic refinement encourages distinct abnormal-cell-related and non-object feature roles, and spatial grouping promotes coarse-support diversity in the initial query pool. Ordered Center Alignment and Quality Coupling (OCAQ) then aligns matched queries with ordered class centers and transfers center consistency to localization-aware classification quality. On the five-class TCT-MS benchmark, CytoSet achieves 39.08 ± 0.31 mAP, outperforming RT-DETR by 1.92 points. After localization-controlled matching, it reduces the severe cross-grade error rate from 7.01 ± 0.29 to 5.78 ± 0.24. In exploratory detector-derived whole-slide image (WSI) analysis, high-risk downgrading decreases from 20.3% to 14.5% on the internal cohort. These results show that conditioning detector queries before Bethesda-order-aware supervision improves fine-grained cytological detection and detector-derived high-grade evidence retention in exploratory WSI analysis.**

***Index Terms*— Bethesda system, cervical cytology, ordered representation learning, object detection.**

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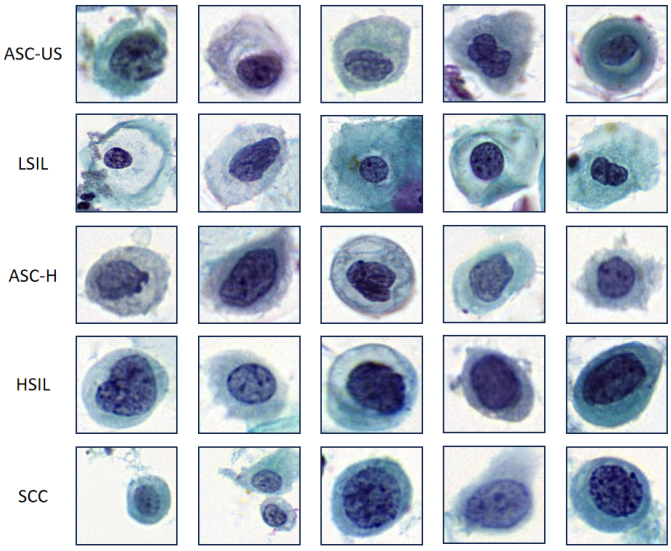


Fig. 1. Representative TCT cell examples from the five squamous Bethesda categories, ordered as ASC-US, LSIL, ASC-H, HSIL, and SCC. The matrix shows substantial intra-class variation, adjacent-grade similarity, and weak adjacent-grade boundaries in fine-grained cervical cytology cell detection.

I. Introduction

CERVICAL cytology is a central component of cervical cancer screening, and the ThinPrep Cytologic Test (TCT) is widely used to identify abnormal cells and precancerous lesions under the Bethesda system [1]. In clinical practice, cytopathologists inspect gigapixel whole-slide images (WSIs) to find diagnostically relevant abnormal cells. This process is time-consuming and subject to inter-observer variability, especially when abnormal cells are sparse, overlapping, morphologically ambiguous, or mixed with complex cytological background [2]–[5]. Automated cell-level detection can therefore localize suspicious cells and provide interpretable cellular evidence for WSI-level Bethesda review.

Slide-level and cell-level systems address complementary aspects of cervical cytology analysis. Many recent cervical cytology AI systems and broader pathology WSI models target slide-level endpoints, such as negative/positive classification, abnormal-cell screening, or coarse risk stratification [6]–[12]. These models provide a global slide assessment, whereas cell-level detection localizes and grades individual abnormalities that can be reviewed within positive slides. This work focuses on the latter task: detecting abnormal squamous cells and

assigning fine-grained Bethesda categories as traceable cellular evidence for downstream review.

This cell-level task poses requirements beyond generic object detection. For this study, we use the operational Bethesda reporting order ASC-US, LSIL, ASC-H, HSIL, and SCC [1]; ASC-H is placed between LSIL and HSIL because it denotes atypia for which HSIL cannot be excluded without meeting the criteria for a definitive HSIL interpretation. Rank distance therefore counts crossed reporting boundaries rather than equal clinical-risk or biological-progression intervals. Standard detectors instead treat these categories as unordered labels and do not distinguish the direction or extent of cross-grade errors. Morphological overlap makes adjacent-grade confusion frequent, whereas errors spanning several reporting boundaries represent larger ordinal deviations; high-to-low errors may additionally reduce the high-grade cellular evidence available for review. As shown in Fig. 1, weak inter-category boundaries coexist with intra-category variation. Consequently, higher aggregate detection accuracy does not necessarily imply more consistent Bethesda assignment after localization.

DETR-like detectors provide a natural query-level interface for imposing ordered supervision on one-to-one matched object representations [13]. Unlike image-level classification, detection applies category supervision only after candidate generation, query initialization, and bipartite matching. The effectiveness of ordered supervision therefore depends on the semantic reliability and spatial coverage of the upstream query pool: semantically unreliable or spatially concentrated candidates can limit the object representations that reach matching. We accordingly condition the query set before applying Bethesda-order-aware learning.

Built upon RT-DETR [14], CytoSet follows a sequential design from query-set conditioning to Bethesda-order-aware query learning. Before decoding, Dual-Codebook Semantic Refinement (DCSR) refines high-level features through weakly supervised dual codebooks, and Spatial-Grouped Query Selector (SGQS) introduces a spatial diversity prior during query initialization. Ordered Center Alignment and Quality Coupling (OCAQ), the core task-specific component, then organizes matched queries around ordered Bethesda class centers and transfers center consistency to localization-aware classification quality. We validate CytoSet through cell-level detection, localization-controlled Bethesda consistency, and WSI-level evidence analysis.

The main contributions are summarized as follows:

- We formulate fine-grained squamous TCT cell detection as Bethesda-order-aware set prediction and evaluate it with localization-controlled metrics that distinguish the extent and direction of cross-grade errors after successful localization.
- We propose a two-stage framework that conditions high-level semantics and coarse-support diversity in the initial query set before applying its core Bethesda-order-aware query learning through OCAQ.
- We validate the framework through cell-level detection, localization-controlled Bethesda consistency, cross-cohort and external evaluation, and exploratory detector-derived WSI evidence analysis.

II. Related Work

A. Automated Cervical Cell Detection

Automated cervical cytology analysis has evolved from morphology-based segmentation of overlapping Pap smear cells [3], [4] to deep learning systems for TCT patches cropped from WSIs [15]–[21]. These systems include CNN-based models, feature-pyramid detectors, YOLO-style detectors, and transformer architectures, and have substantially advanced abnormal-cell localization and recognition. Recent large-scale and hierarchical datasets further highlight the fine-grained structure of cervical cell classification [22]. However, evaluations remain dominated by AP, accuracy, or coarse abnormality recognition, and most detectors retain unordered category supervision [23], [24]. Existing results therefore do not establish whether improved detection also yields more order-consistent Bethesda assignment after localization.

B. Query Selection in DETR-like Detectors

DETR-like detectors formulate object detection as end-to-end set prediction with one-to-one Hungarian assignment [13]. Later variants improve proposal generation, convergence, and decoder refinement through deformable attention, encoder-generated proposals, dense multi-scale supervision, denoising training, and query refinement or pruning [14], [25]–[29]. At the proposal-to-query interface, we focus on cross-scale candidates that share coarse spatial support and may become over-represented in global ranking. Because Hungarian assignment is applied only after query initialization, one-to-one matching does not prevent several high-ranked encoder proposals from concentrating on the same local evidence. The issue addressed by SGQS is therefore candidate-pool over-concentration before matching, rather than duplicate assignment itself.

C. Fine-grained Representation and Prototype Learning

Fine-grained recognition methods, including metric learning, prototype learning, angular-margin learning, structure-aware nuclear representation learning, and order-aware grading, have been used to address small inter-class differences, ordered labels, and large intra-class variation [10], [30]–[39]. Learned codebooks provide a complementary means of reconstructing structured feature patterns [40], [41]. Prototype- and center-based objectives encourage compact class representations, while ordinal methods distinguish errors according to label order or distance. Conventional ordinal objectives primarily reweight output errors or structure image-level embeddings. Detection introduces two additional interfaces: category supervision acts on one-to-one matched object queries, and proposal confidence also estimates localization quality. An order-aware detector must therefore connect representation geometry to the quality target used for query selection. DCSR uses weak region supervision to encourage distinct abnormal-cell-related and non-object codebook roles, whereas OCAQ structures matched decoder queries and couples center consistency to quality-aware set prediction.

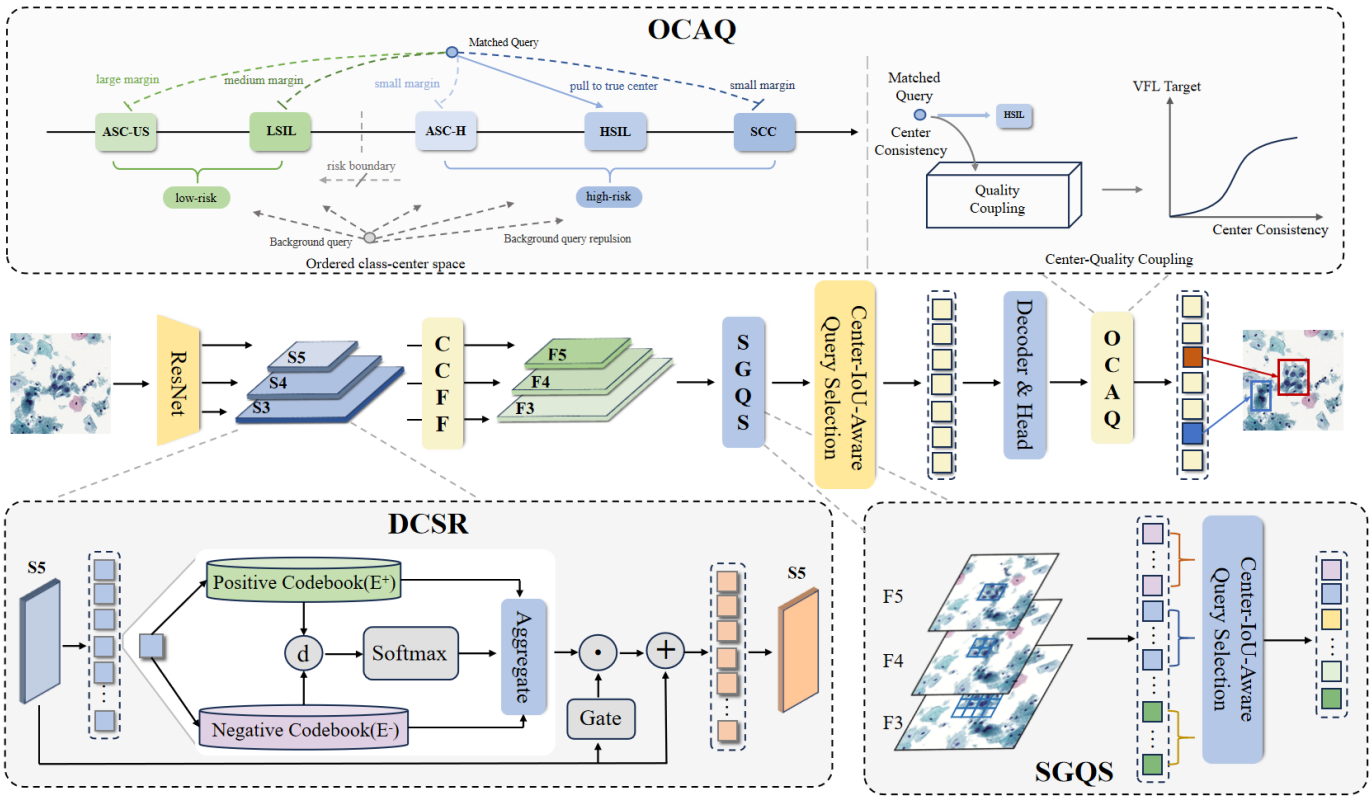


Fig. 2. Overall architecture of CytoSet. DCSR and SGQS condition the semantic reliability and spatial diversity of the query set before decoding, after which OCAQ aligns matched queries in an ordered Bethesda center space and couples center consistency with classification quality.

III. METHOD

A. Overall Architecture

Figure 2 shows the overall framework of CytoSet. Built upon RT-DETR [14], CytoSet organizes its adaptations into two sequential levels: reliable query-set construction before decoding and Bethesda-order-aware learning of matched decoder queries. The original Hungarian matching cost and box-regression pathway remain unchanged; the center-aware quality target affects proposal ranking and supervised classification rather than bipartite assignment or box regression.

Given an input patch, the backbone extracts three feature levels, denoted as $S3$, $S4$, and $S5$. DCSR refines projected $S5$ before the cross-scale feature fusion module (CCFF) fuses it with $S3$ and $S4$ into the multi-scale encoder memory. This places semantic refinement before cross-scale aggregation while leaving the lower-level branches unchanged.

On top of the encoder memory, the proposal head predicts candidate scores at all spatial locations. Instead of directly selecting global top-K candidates, SGQS groups proposals sharing the same coarse $S5$ spatial support, retains at most one representative per group, and performs top-K selection on the reduced set. Together, DCSR and SGQS condition the semantic reliability and spatial diversity of the query set entering the decoder. OCAQ then aligns the resulting matched queries in a normalized ordered Bethesda representation space. Its Center-IoU-Aware Query Selection (CIQS) extends the IoU-aware query selection of RT-DETR by incorporating query–center consistency into the quality target used to rank encoder proposals.

B. Dual-Codebook Semantic Refinement

Dual-Codebook Semantic Refinement (DCSR) is an encoder-side semantic adaptation module applied to projected $S5$. It introduces two jointly assigned codebooks whose roles are weakly supervised: the positive branch is encouraged to capture abnormal-cell-related patterns, whereas the negative branch is encouraged to capture non-object patterns. Their differentiable reconstruction refines the main feature stream, with auxiliary commitment, utilization, and separation objectives used to discourage degenerate codebook use.

For each projected token $\mathbf{x}_i \in \mathbb{R}^C$, DCSR computes $\mathbf{u}_i = W_{\text{in}}\mathbf{x}_i$ and $\bar{\mathbf{u}}_i = \text{Norm}(\mathbf{u}_i)$. The positive and negative codebooks are concatenated as $\mathbf{E} = [\mathbf{E}^+; \mathbf{E}^-]$, with $\bar{\mathbf{e}}_k = \text{Norm}(\mathbf{e}_k)$. Each token is jointly assigned over all $K_+ + K_-$ codewords using $d_{i,k} = 2 - 2\langle \bar{\mathbf{u}}_i, \bar{\mathbf{e}}_k \rangle$ and $a_{i,k} = \exp(-d_{i,k}/\tau) / \sum_{j=1}^{K_+ + K_-} \exp(-d_{i,j}/\tau)$. The reconstruction $\mathbf{r}_i = \sum_{k=1}^{K_+ + K_-} a_{i,k}\bar{\mathbf{e}}_k$ is injected through a gated residual path:

$$\mathbf{y}_i = \mathbf{x}_i + g(\mathbf{x}_i) \odot W_{\text{out}}\mathbf{r}_i. \tag{1}$$

This residual design preserves the original feature stream while allowing codebook-guided correction. All spatial tokens attend jointly to both codebooks; region labels are used only by the auxiliary separation loss during training.

Let Ω^+ and Ω^- denote central object-core and non-object token sets. The object core reduces ambiguity from overlap-

ping cytoplasm and imprecise box edges; non-object tokens lie outside all annotated boxes, and remaining tokens are ignored by $\mathcal{L}_{\text{sep}}$. Using $d_{i,k}^{\text{sep}} = 2 - 2\langle \text{sg}(\bar{\mathbf{u}}_i), \bar{\mathbf{e}}_k \rangle$, let d_i^{++} and d_i^{+-} be the nearest distances from a positive token to the positive and negative codebooks, with d_j^{--} and d_j^{-+} defined analogously for a non-object token. We set $\mathcal{L}_{\text{sep}} = \mathcal{L}_{\text{pull}} + \mathcal{L}_{\text{push}}$:

$$\mathcal{L}_{\text{pull}} = \frac{1}{|\Omega^+|} \sum_{i \in \Omega^+} d_i^{++} + \frac{1}{|\Omega^-|} \sum_{j \in \Omega^-} d_j^{--}. \tag{2}$$

$$\begin{aligned} \mathcal{L}_{\text{push}} = &\frac{1}{|\Omega^+|} \sum_{i \in \Omega^+} \left[\beta d_i^{++} + \delta - d_i^{+-} \right]_+ \\ &+ \frac{1}{|\Omega^-|} \sum_{j \in \Omega^-} \left[\beta d_j^{--} + \delta - d_j^{-+} \right]_+ . \end{aligned} \tag{3}$$

Following vector-quantized representation learning [40], let $\mathcal{S}$ denote all spatial tokens and $\text{sg}(\cdot)$ denote stop-gradient. VQ-style detached targets are used for commitment and utilization balancing while the reconstruction path remains differentiable. Specifically, $\mathbf{r}_i^{\text{cb}} = \sum_k \text{sg}(a_{i,k}) \bar{\mathbf{e}}_k$. For utilization balancing, $a_{i,k}^{\text{div}}$ denotes the same soft assignment recomputed with $\text{sg}(\bar{\mathbf{u}}_i)$. Let $\mathcal{K}_b$ index branch $b \in \{+, -\}$. We define $\tilde{p}_k^b = |\mathcal{S}|^{-1} \sum_{i \in \mathcal{S}} a_{i,k}^{\text{div}}$ and $\bar{p}_k^b = (\tilde{p}_k^b + \varepsilon) / \sum_{j \in \mathcal{K}_b} (\tilde{p}_j^b + \varepsilon)$. The complete DCSR objective is

$$\begin{aligned} \mathcal{L}_{\text{cb}} &= \frac{1}{|\mathcal{S}|} \sum_i \| \text{sg}(\bar{\mathbf{u}}_i) - \mathbf{r}_i^{\text{cb}} \|_2^2, \\ \mathcal{L}_{\text{com}} &= \frac{1}{|\mathcal{S}|} \sum_i \| \bar{\mathbf{u}}_i - \text{sg}(\mathbf{r}_i) \|_2^2, \\ \mathcal{L}_{\text{div}} &= \sum_{b \in \{+,-\}} \sum_{k \in \mathcal{K}_b} \bar{p}_k^b \log(K_b \bar{p}_k^b + \varepsilon), \\ \mathcal{L}_{\text{DCSR}} &= \mathcal{L}_{\text{cb}} + \lambda_{\text{com}} \mathcal{L}_{\text{com}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{sep}} \mathcal{L}_{\text{sep}}. \end{aligned} \tag{4}$$

The commitment terms couple tokens and codewords, $\mathcal{L}_{\text{div}}$ balances codeword use within each branch, and $\mathcal{L}_{\text{sep}}$ is the only region-supervised term. The reconstruction remains jointly assigned across both codebooks and differentiable to the detection objective.

C. Spatial-Grouped Query Selector

In set prediction, multi-scale candidates sharing the same coarse spatial support can enter the global ranking independently and become over-represented in the initial query pool. Spatial-Grouped Query Selector (SGQS) imposes a spatial diversity prior by introducing local cross-scale competition before global top-K selection, allowing each coarse group to contribute one representative.

For each coarse location u on the $S5$ grid, SGQS collects the associated candidates from three feature scales:

$$\mathcal{G}_u = \{(5, u)\} \cup \{(4, v) \mid v \in \Phi_4(u)\} \cup \{(3, v) \mid v \in \Phi_3(u)\}, \tag{5}$$

where $\Phi_4(u)$ and $\Phi_3(u)$ contain the four $S4$ and sixteen $S3$ locations mapped to the same coarse support, respectively. Thus, each group contains one $S5$, four $S4$, and sixteen $S3$ candidates. For each candidate (s, v), the proposal score is $\rho_{s,v} = \max_c \ell_{s,v,c}$, where $\ell_{s,v,c}$ denotes the encoder proposal confidence supervised by the center-IoU-aware quality target. SGQS keeps the highest-scoring representative $g_u^\star = \arg\max_{(s,v) \in \mathcal{G}_u} \rho_{s,v}$ and applies score-guided top-K selection on the reduced set; hence, both within-group maximization and global ranking follow CIQS:

$$\mathcal{Q}_0 = \text{TopK}_K(\{\rho(g_u^\star)\}_{u \in S5}). \tag{6}$$

SGQS changes the candidate pool rather than the top-K rule. Restricting each coarse spatial group to one representative increases coarse-support diversity in the reduced pool. It supplies a spatial-coverage prior for query initialization rather than ordinal supervision.

D. Ordered Center Alignment and Quality Coupling

Ordered Center Alignment and Quality Coupling (OCAQ) incorporates the operational order used in this study, ASC-US < LSIL < ASC-H < HSIL < SCC [1], into matched decoder-query supervision while retaining standard five-class outputs. The distance $|r(a) - r(b)|$ counts crossed reporting boundaries rather than equal clinical, morphological, or biological spacing.

OCAQ has two core blocks. Bethesda-ordered query alignment organizes matched queries around category centers and includes a one-way penalty for high-to-low assignments. Center-quality coupling then transfers center consistency to the detector's IoU-aware classification target. Category geometry and background separation serve as supporting regularization of the shared center space.

1) Bethesda-Ordered Query Alignment: Let $\mathcal{M}$ denote the index set of matched foreground queries. For supervised decoder layer ℓ, OCAQ obtains $\mathbf{z}_i^{(\ell)} = \text{Norm}(h_\phi(\mathbf{q}_i^{(\ell)})) \in \mathbb{R}^{64}$; the layer superscript is omitted below. Each class maintains a shared normalized EMA center, initialized when that class is first observed. If $\bar{\mathbf{z}}_c$ is the mean final-layer embedding of current matched queries with label c, the center is updated as $\boldsymbol{\mu}_c \leftarrow \text{Norm}(m\boldsymbol{\mu}_c + (1-m)\bar{\mathbf{z}}_c)$. Let $\mathcal{C}_t$ denote the centers available at iteration t and $\mathcal{M}_t = \{i \in \mathcal{M} : y_i \in \mathcal{C}_t\}$. The projection and centers are shared across supervised decoder layers, whereas centers are updated only from final-layer matches.

The five categories have ranks $r(c) \in \{0, 1, 2, 3, 4\}$. For $c \in \mathcal{C}_t$, the similarity is $s_{i,c} = \langle \mathbf{z}_i, \boldsymbol{\mu}_c \rangle$, and $p_{i,c} = \text{softmax}_{c \in \mathcal{C}_t}(s_{i,c}/\tau_c)$. This center-space posterior is distinct from the detector classification posterior; they are coupled through the shared decoder representation and the center-modulated IoU-aware Varifocal Loss (VFL) target.

At the query level, OCAQ aligns each valid matched foreground query in $\mathcal{M}_t$ with the ordered category-center space using a rank-distance-aware margin:

$$\begin{aligned} \mathcal{L}_{\text{ord}} = &\frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \Bigg[(1 - s_{i,y_i}) + \frac{1}{\max(1, |\mathcal{C}_t| - 1)} \sum_{c \in \mathcal{C}_t \setminus \{y_i\}} \\ &\left[m_0 + \frac{|r(y_i) - r(c)|}{R} - s_{i,y_i} + s_{i,c} \right]_+ \Bigg], \end{aligned} \tag{7}$$

where $R = 4$. The first term attracts the query to its correct center, while the second increases the separation margin with the number of crossed category boundaries.

The margin-based objective constrains pairwise query-to-center relations. We additionally measure the expected crossed-boundary cost under the center-space posterior:

$$\mathcal{L}_{\mathrm{cal}} = \frac{1}{|\mathcal{M}_t|} \sum_{i \in \mathcal{M}_t} \sum_{c \in \mathcal{C}_t} p_{i,c} \left| r(c) - r(y_i) \right|. \tag{8}$$

$\mathcal{L}_{\mathrm{ord}}$ defines rank-dependent pairwise alignment, whereas $\mathcal{L}_{\mathrm{cal}}$ applies the same crossed-boundary count to the full center posterior. Because the rank gap is absolute, posterior mass on either side of the target contributes positively rather than canceling as it could under a signed rank residual. Thus, $\mathcal{L}_{\mathrm{cal}}$ complements the pairwise margins by constraining the complete center posterior.

Because high-to-low errors can reduce the high-grade cellular evidence available under the task definition, query alignment is complemented by direction-specific supervision for matched high-risk queries. Let $\mathcal{C}_{\mathrm{H}} = \{\mathrm{ASC\text{-}H}, \mathrm{HSIL}, \mathrm{SCC}\}$ and $\mathcal{C}_{\mathrm{L}} = \{\mathrm{ASC\text{-}US}, \mathrm{LSIL}\}$ denote the high- and low-risk sets, and let $\mathcal{M}_{\mathrm{H},t} = \{i \in \mathcal{M}_t : y_i \in \mathcal{C}_{\mathrm{H}}\}$. The one-way downgrade regularizer is

$$\mathcal{L}_{\mathrm{down}} = \frac{1}{|\mathcal{M}_{\mathrm{H},t}|} \sum_{i \in \mathcal{M}_{\mathrm{H},t}} \left[\max_{c \in \mathcal{C}_{\mathrm{L}} \cap \mathcal{C}_t} s_{i,c} - \max_{c \in \mathcal{C}_{\mathrm{H}} \cap \mathcal{C}_t} s_{i,c} \right]_+ . \tag{9}$$

This term penalizes a high-risk query when its maximum low-risk-center similarity exceeds its maximum high-risk-center similarity.

2) Center-IoU-Aware Query Selection and Quality Coupling: RT-DETR performs IoU-aware query selection by ranking encoder proposals with classification scores whose positive targets incorporate localization quality [14]. This favors proposals that are both discriminative and well localized before top-K initialization, but it does not account for their consistency with the ordered category representation. We extend this mechanism through CIQS, using center consistency to modulate the IoU-aware quality target that supervises the proposal ranking score. For a matched foreground candidate, let $u_i = \mathrm{IoU}(\hat{\mathbf{b}}_i, \mathbf{b}_i)$ and $c_i = (1 + s_{i,y_i})/2$. The center-IoU-aware positive target is

$$q_i^{\mathrm{joint}} = \mathrm{sg}\left[u_i \big((1-\eta) + \eta c_i \big) \right], \qquad \eta \in [0,1]. \tag{10}$$

The joint target is stop-gradient. Because $q_i^{\mathrm{joint}} \le u_i$, IoU remains an upper bound on the positive classification target; center consistency can lower, but not raise, this target relative to u_i. The same target supervises the proposal and decoder classification terms.

Two auxiliary regularizers stabilize the shared center space. Query-to-center alignment does not explicitly constrain the relative geometry among category representations, so we impose an anchor-relative monotonic relation at the category level. At each supervised decoder layer, let $\mathcal{M}_c = \{i \in \mathcal{M}_t : y_i = c\}$ and $\mathcal{C}_B = \{c : |\mathcal{M}_c| > 0\}$. The differentiable category representative is

$$\mathbf{v}_c = \mathrm{Norm}\left(\frac{1}{|\mathcal{M}_c|} \sum_{i \in \mathcal{M}_c} \mathbf{z}_i \right), \qquad c \in \mathcal{C}_B. \tag{11}$$

Define $d_{ab} = (1 - \langle \mathbf{v}_a, \mathbf{v}_b \rangle)/2$, $\Delta_{ab} = |r(a) - r(b)|$, and $\mathcal{T}_B = \{(a,b,c) \in \mathcal{C}_B^3 : a,b,c \text{ are distinct}, \Delta_{ab} < \Delta_{ac}\}$. The geometry objective is

$$\mathcal{L}_{\mathrm{geo}} = \frac{1}{|\mathcal{T}_B|} \sum_{(a,b,c) \in \mathcal{T}_B} \left[d_{ab} - d_{ac} + m_{\mathrm{geo}} \frac{\Delta_{ac} - \Delta_{ab}}{R} \right]_+ . \tag{12}$$

For a fixed anchor, $\mathcal{L}_{\mathrm{geo}}$ encourages a smaller representation distance for a smaller reporting-rank gap.

Unmatched queries $\mathcal{U}$ are additionally repelled from the foreground Bethesda center space:

$$\mathcal{L}_{\mathrm{bg}} = \frac{1}{|\mathcal{U}|} \sum_{i \in \mathcal{U}} \left[\max_{c \in \mathcal{C}_t} s_{i,c} - \gamma_{\mathrm{bg}} \right]_+ . \tag{13}$$

E. Loss Function

The training objective retains the RT-DETR box losses and Hungarian assignment, replaces the positive VFL target in supervised classification outputs with q_i^{joint}, and adds DCSR and OCAQ regularization. The detection loss is

$$\mathcal{L}_{\mathrm{det}} = \lambda_{\mathrm{vfl}} \mathcal{L}_{\mathrm{vfl}}^{\mathrm{joint}} + \lambda_{\mathrm{box}} \mathcal{L}_1 + \lambda_{\mathrm{giou}} \mathcal{L}_{\mathrm{giou}} + \sum_{\ell \in \mathcal{A}} \mathcal{L}_{\mathrm{aux}}^{(\ell)}, \tag{14}$$

where $\mathcal{L}_{\mathrm{vfl}}^{\mathrm{joint}}$ is supervised by q_i^{joint} using Varifocal Loss [42], $\mathcal{L}_1$ and $\mathcal{L}_{\mathrm{giou}}$ are box regression losses, and $\mathcal{A}$ denotes the ordinary auxiliary decoder outputs. The classification term of each supervised auxiliary decoder layer uses the same center-modulated positive target.

The OCAQ regularization is

$$\begin{aligned} \mathcal{L}_{\mathrm{OCAQ}} = {} & \lambda_{\mathrm{ord}} \mathcal{L}_{\mathrm{ord}} + \lambda_{\mathrm{cal}} \mathcal{L}_{\mathrm{cal}} + \lambda_{\mathrm{down}} \mathcal{L}_{\mathrm{down}} \\ & + \lambda_{\mathrm{geo}} \mathcal{L}_{\mathrm{geo}} + \lambda_{\mathrm{bg}} \mathcal{L}_{\mathrm{bg}}. \end{aligned} \tag{15}$$

It aggregates ordered alignment, asymmetric downgrade control, and supporting center-space regularization, and is summed over the supervised decoder layers; quality coupling enters the detector classification term through q_i^{joint}. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{det}} + \lambda_{\mathrm{dcsr}} \mathcal{L}_{\mathrm{DCSR}} + \mathcal{L}_{\mathrm{OCAQ}}. \tag{16}$$

IV. Experiments

To evaluate Bethesda-order-aware set prediction from cell-level detection to downstream evidence retention, we assess CytoSet at three progressive levels using a multi-scanner internal benchmark and two external datasets. First, **cell-level detection** measures standard localization and recognition accuracy against generic detectors. Second, **localization-controlled Bethesda consistency** conditions grade-error analysis on successful localization to evaluate severe grade drift, mean ordinal error, and high-risk downgrading. Finally, an exploratory **detector-derived WSI evidence-retention analysis** examines whether improved cell-level detection preserves high-grade cellular evidence for downstream slide review.

A. Ethics Statement

This study was approved by the Ethics Committee of Fujian Maternity and Child Health Hospital (approval No. 2023KY045; April 27, 2023), with the requirement for informed consent waived by the committee.

B. Implementation Details

All images are resized to 640×640. CytoSet and RT-DETR use the official RT-DETR-R50vd $6\times$ configuration and the same pretrained PResNet-50vd backbone. Both are trained with AdamW for 72 epochs (batch size 4; learning rates 10^{-4} and 10^{-5} for the detector and backbone) using the original schedule, EMA, and augmentation recipe; the highest-validation-mAP checkpoint from each run is evaluated. Other detectors retain their official recipes and pretrained initialization when available, except for adapting their heads to five categories. All methods share the TCT-MS patient-level split, input resolution, and evaluation protocol.

DCSR, SGQS, and OCAQ are optimized end-to-end with the RT-DETR objective. DCSR uses 64 codewords per branch, $\tau = 0.1$, $\beta = 1.2$, $\delta = 0.1$, and a central $30\% \times 30\%$ object-core region; $(\lambda_{\text{com}}, \lambda_{\text{div}}, \lambda_{\text{sep}}, \lambda_{\text{dcsr}}) = (0.05, 0.1, 1.0, 0.1)$. OCAQ projects 256-dimensional queries into 64-dimensional normalized embeddings and uses $m = 0.9$, $\tau_c = 0.1$, $\eta = 0.5$, $m_0 = 0.20$, $m_{\text{geo}} = 0.20$, and $\gamma_{\text{bg}} = 0.30$; $(\lambda_{\text{ord}}, \lambda_{\text{cal}}, \lambda_{\text{down}}, \lambda_{\text{geo}}, \lambda_{\text{bg}}) = (0.5, 0.5, 0.2, 0.5, 0.1)$. These values are selected on the TCT-MS validation split and fixed across the comparison, transfer, and ablation experiments.

For the Rank-VFL baseline, we retain the original RT-DETR varifocal loss and weight each non-target class of a matched query by $1 + |r(y_i) - r(c)|/R$; positive and unmatched-query terms remain unchanged. Rank-VFL therefore isolates output-level rank weighting without introducing class centers, query-space geometry, or center-quality coupling.

All experiments are conducted on one NVIDIA Tesla V100-SXM2 GPU with 32 GB memory. Main experiments are repeated with three random seeds, 42, 2026, and 3407, and reported as mean $\pm$ standard deviation. Inference uses a single model without test-time augmentation or ensembling.

C. Datasets and Splits

We construct the main benchmark from two independent internal TCT subsets, denoted as `0506` and `0818`, acquired from different scanners (3DHISTECH Pannoramic SCAN and Jiangfeng KF-PRO-400, respectively) with no case overlap. Both subsets follow ThinPrep preparation with Papanicolaou staining and are scanned at $40\times$ with 0.2438 microns-per-pixel resolution. All patch images are 640×640 pixels. All cell-level annotations were performed by nine cytopathologists and subsequently reviewed by one chief cytopathologist. Disagreements and ambiguous labels were resolved by the chief cytopathologist to ensure annotation consistency. The five foreground squamous categories are ASC-US, LSIL, ASC-H, HSIL, and SCC. For main comparison and ablation studies, the two subsets are merged as TCT-MS. Table I summarizes the dataset scale.

Patient-level splits are created independently for the two subsets with an approximate 7:1.5:1.5 train/validation/test ratio and then merged to avoid slide-level leakage; the patch counts after cropping are reported in Table I. Transfer and external evaluation use Tianchi [43] and CRIC [44]. Tianchi WSIs are acquired at $20\times$, resized by $2\times$ to a $40\times$-equivalent scale, and split by WSI after cropping from 500 positive WSIs. The slides follow ThinPrep preparation with Papanicolaou staining; SCC is retained as an empty category in the five-class label space. CRIC provides conventional Pap smear images with cell-level annotations mapped to the same five squamous categories and is used to evaluate portability across preparation protocols.

TABLE I
DATASET STATISTICS. TIANCHI IS REPORTED AT THE $40\times$-EQUIVALENT SCALE AFTER $2\times$ RESIZING FROM $20\times$ WSIS.

Dataset	Mag.	WSIs	Train img.	Val img.	Test img.	Total img.	Cell boxes
`0506`	40×	214	10,158	1,984	2,056	14,198	14,838
`0818`	40×	231	10,734	1,517	1,595	13,846	16,564
TCT-MS	40×	445	**20,892**	**3,501**	**3,651**	**28,044**	**31,402**
Tianchi	40×	500	4,948	1,030	1,005	6,983	8,635
CRIC	40×	–	2,026	84	84	2,194	4,490

TABLE II
CLASS DISTRIBUTION OF CELL-BOX ANNOTATIONS. HIGH-RISK TOTAL INCLUDES ASC-H, HSIL, AND SCC; TIANCHI CONTAINS NO SCC ANNOTATIONS, WITH SCC RETAINED AS AN EMPTY CATEGORY IN THE FIVE-CLASS LABEL SPACE.

Class	`0506`	`0818`	TCT-MS	Tianchi	CRIC
ASC-US	4,584 (30.9%)	6,041 (36.5%)	10,625 (33.8%)	2,108 (24.4%)	1,479 (32.9%)
LSIL	2,146 (14.5%)	5,273 (31.8%)	7,419 (23.6%)	876 (10.1%)	1,715 (38.2%)
ASC-H	1,300 (8.8%)	3,702 (22.3%)	5,002 (15.9%)	1,057 (12.2%)	402 (9.0%)
HSIL	5,927 (39.9%)	1,126 (6.8%)	7,053 (22.5%)	4,594 (53.2%)	663 (14.8%)
SCC	881 (5.9%)	422 (2.5%)	1,303 (4.1%)	0 (0.0%)	231 (5.1%)
High-risk total	8,108 (54.6%)	5,250 (31.7%)	13,358 (42.5%)	5,651 (65.4%)	1,296 (28.9%)

D. Evaluation Metrics

We follow the COCO protocol [51] and report mAP@[0.5:0.95], AP@0.5, and AP@0.75. Main TCT-MS comparisons and ablations are reported as mean $\pm$ std over three random seeds. We also provide class-wise, confusion-based, and WSI-level evidence analyses.

To evaluate Bethesda consistency after successful localization, we compute localization-controlled metrics using class-agnostic matching. For each image, predictions are sorted by confidence and greedily matched one-to-one to ground-truth boxes by maximum IoU, with the default threshold $\theta = 0.50$. Let $\mathcal{M}_\theta = \{(g, p)\}$ denote the matched ground-truth–prediction pairs. The categories ASC-US, LSIL, ASC-H, HSIL, and SCC are assigned ordinal ranks from 0 to 4. For a matched pair (g, p) with labels y_g and y_p, $d(g, p) = |r(y_g) - r(y_p)|$ counts the crossed category boundaries.

These metrics condition Bethesda-grade evaluation on successful class-agnostic localization, reducing the confounding effect of missed localization:

$$\text{LC-Acc} = \frac{1}{|\mathcal{M}_\theta|} \sum_{(g,p)\in\mathcal{M}_\theta} \mathbf{1}[y_g = y_p], \tag{17}$$

$$\text{SCR} = \frac{1}{|\mathcal{M}_\theta|} \sum_{(g,p)\in\mathcal{M}_\theta} \mathbf{1}[d(g, p) \geq 2], \tag{18}$$

$$\text{MOE} = \frac{1}{|\mathcal{M}_\theta|} \sum_{(g,p)\in\mathcal{M}_\theta} d(g, p). \tag{19}$$

Here, LC-Acc denotes localization-controlled accuracy, SCR denotes the severe cross-grade error rate, and MOE denotes mean ordinal error.

TABLE III
MAIN COMPARISON ON TCT-MS. VALUES ARE MEAN ± STD OVER THREE RANDOM SEEDS. BETHESDA CONSISTENCY METRICS ARE COMPUTED USING CLASS-AGNOSTIC MATCHES AT IOU≥0.5. ALL METRICS EXCEPT MOE ARE PERCENTAGES; MOE IS IN ORDINAL-RANK UNITS. BOLD DENOTES THE BEST RESULT.

Method	Backbone	mAP↑	AP@0.5↑	AP@0.75↑	LC-Acc↑	SCR↓	MOE↓	HR-Recall↑	HR-DG↓
Faster R-CNN [45]	ResNet-50	35.10±0.78	48.64±0.72	39.36±0.83	68.72±0.95	8.84±0.52	0.446±0.015	85.06±1.38	13.32±0.96
RetinaNet [46]	ResNet-50	33.86±0.88	47.82±0.91	37.28±0.92	66.42±1.18	9.40±0.64	0.482±0.019	83.56±1.76	14.74±1.12
Sparse R-CNN [47]	ResNet-50	35.32±0.70	48.10±0.74	39.68±0.80	69.18±0.90	8.78±0.45	0.437±0.013	84.72±1.31	13.10±0.88
YOLOv5-M [48]	–	35.78±0.59	49.94±0.60	39.42±0.66	69.10±0.72	8.42±0.39	0.430±0.011	86.02±1.10	12.78±0.78
YOLOv8-M [49]	–	36.28±0.52	50.46±0.54	40.12±0.61	70.08±0.65	8.16±0.38	0.418±0.010	86.38±1.03	12.22±0.72
YOLO11-M [50]	–	36.52±0.50	50.74±0.56	40.66±0.57	69.82±0.69	8.08±0.36	0.421±0.010	85.96±0.98	12.34±0.69
DETR [13]	ResNet-50	31.12±1.10	43.18±1.02	34.42±1.20	63.46±1.34	10.84±0.74	0.534±0.023	81.48±2.12	16.08±1.26
Deformable DETR [25]	ResNet-50	35.82±0.64	49.20±0.68	40.02±0.75	69.42±0.82	8.48±0.43	0.434±0.012	85.34±1.24	12.86±0.86
DINO [26]	ResNet-50	36.74±0.45	50.62±0.50	41.08±0.55	72.86±0.56	7.18±0.33	0.378±0.010	87.62±0.94	10.82±0.66
RT-DETR [14]	PResNet-50vd	37.16±0.36	51.18±0.40	41.58±0.47	74.39±0.44	7.01±0.29	0.360±0.008	88.62±0.82	10.37±0.72
RT-DETR + Rank-VFL	PResNet-50vd	37.43±0.39	51.55±0.43	41.82±0.50	75.02±0.48	6.69±0.31	0.349±0.009	88.95±0.79	9.61±0.68
CytoSet	PResNet-50vd	**39.08±0.31**	**53.72±0.36**	**43.52±0.35**	**76.62±0.42**	**5.78±0.24**	**0.316±0.009**	**89.86±0.64**	**7.82±0.46**

TABLE IV
PER-CLASS AP COMPARISON AT IOU THRESHOLDS OF 0.5 AND 0.75.

Class	RT-DETR		CytoSet	
	AP@0.5	**AP@0.75**	**AP@0.5**	**AP@0.75**
ASC-US	65.6±0.35	58.8±0.42	**66.3±0.28**	**59.1±0.31**
LSIL	67.5±0.43	54.6±0.50	**71.3±0.34**	**57.2±0.39**
ASC-H	34.7±0.56	25.3±0.63	**36.8±0.47**	**26.4±0.52**
HSIL	70.9±0.31	63.2±0.38	**74.0±0.27**	**64.5±0.32**
SCC	17.3±0.65	6.1±0.70	**20.1±0.56**	**10.3±0.62**

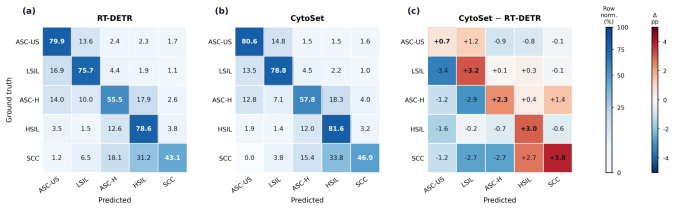


Fig. 3. Localization-controlled confusion analysis. (a) RT-DETR. (b) CytoSet. (c) CytoSet–RT-DETR difference in percentage points. Distant off-diagonal cells indicate severe cross-grade errors.

We further define ASC-H, HSIL, and SCC as high-risk classes, and ASC-US and LSIL as low-risk classes. Let $\mathcal{H}$ and $\mathcal{L}$ denote the high- and low-risk label sets, $\mathcal{G}_{\mathcal{H}}$ all high-risk ground-truth cells, and $\mathcal{M}_{\mathcal{H}} = \{(g,p) \in \mathcal{M}_\theta : y_g \in \mathcal{H}\}$ matched high-risk cells. The high-risk recall (HR-Recall) and high-risk downgrading rate (HR-DG) are:

$$\text{HR-Recall} = \frac{1}{|\mathcal{G}_{\mathcal{H}}|} \sum_{g \in \mathcal{G}_{\mathcal{H}}} \mathbf{1}[\exists p : (g,p) \in \mathcal{M}_\theta,\ y_p \in \mathcal{H}], \quad (20)$$

$$\text{HR-DG} = \frac{1}{|\mathcal{M}_{\mathcal{H}}|} \sum_{(g,p) \in \mathcal{M}_{\mathcal{H}}} \mathbf{1}[y_p \in \mathcal{L}]. \quad (21)$$

HR-Recall measures whether high-risk cells are both localized and retained within the high-risk group, whereas HR-DG measures high-risk-to-low-risk downgrading after localization succeeds. SCR, MOE, and HR-DG are conditional on successful class-agnostic localization and should therefore be read together with HR-Recall, whose denominator includes all high-risk ground-truth cells and remains sensitive to missed detections.

E. Main Results on TCT-MS

Table III compares standard detection and localization-controlled Bethesda metrics, with Rank-VFL included as a simple rank-aware baseline.

CytoSet achieved the highest performance among the compared methods. Compared with RT-DETR, it improved mAP, AP@0.5, and AP@0.75 by 1.92, 2.54, and 1.94 points, respectively, reaching 39.08 ± 0.31 mAP, 53.72 ± 0.36 AP@0.5, and 43.52 ± 0.35 AP@0.75. These results indicate improved abnormal-cell localization and fine-grained category recognition on TCT-MS.

CytoSet also reduced grade-level errors after localization. Compared with RT-DETR, SCR decreased from 7.01 ± 0.29 to 5.78 ± 0.24, MOE from 0.360 ± 0.008 to 0.316 ± 0.009, and HR-DG from 10.37 ± 0.72 to 7.82 ± 0.46. Rank-VFL provided smaller improvements, reaching 6.69 ± 0.31 SCR, 0.349 ± 0.009 MOE, and 9.61 ± 0.68 HR-DG. Together, these results show that simple rank weighting is beneficial, while CytoSet provides larger gains in both detection performance and Bethesda consistency.

1) Category-wise Analysis: Table IV reports per-class AP@0.5 and AP@0.75. CytoSet improves both metrics across all five categories, with clear gains on LSIL, HSIL, and SCC. SCC, which has the fewest annotations in TCT-MS, remains the most challenging category, but its AP@0.75 increases from 6.1 to 10.3 under CytoSet.

2) Localization-Controlled Confusion Analysis: Under the class-agnostic matching protocol, Fig. 3 shows a more concentrated diagonal for CytoSet and reduced off-diagonal mass across multiple Bethesda grades and the low-/high-risk boundary, consistent with the lower SCR, MOE, and HR-DG in Table III.

Figure 4 presents qualitative detection results of CytoSet and competing methods on representative TCT patches.

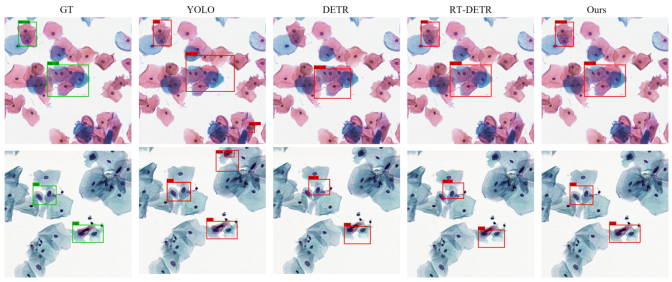


Fig. 4. Qualitative detection results on representative TCT patches. Green boxes denote ground-truth annotations and red boxes denote model predictions.

TABLE V
ABLATION STUDY OF DCSR, SGQS, AND OCAQ. ALL COMBINATIONS ARE EVALUATED WITH STANDARD DETECTION METRICS AND KEY LOCALIZATION-CONTROLLED BETHESDA CONSISTENCY METRICS. VALUES ARE MEAN ± STD OVER THREE RANDOM SEEDS. ALL METRICS EXCEPT MOE ARE PERCENTAGES; MOE IS IN ORDINAL-RANK UNITS.

DCSR	SGQS	OCAQ	mAP↑	AP@0.5↑	AP@0.75↑	SCR↓	MOE↓	HR-DG↓
×	×	×	37.16±0.36	51.18±0.40	41.58±0.47	7.01±0.29	0.360±0.008	10.37±0.72
✓	×	×	38.08±0.44	52.74±0.48	42.48±0.51	6.92±0.34	0.355±0.011	10.18±0.78
×	✓	×	37.82±0.46	52.06±0.51	42.76±0.52	7.06±0.36	0.362±0.012	10.44±0.82
×	×	✓	37.68±0.42	51.84±0.47	42.12±0.49	6.42±0.30	0.339±0.010	8.94±0.64
✓	✓	×	38.88±0.39	53.52±0.43	43.23±0.45	6.86±0.33	0.351±0.011	10.02±0.75
✓	×	✓	38.50±0.38	52.82±0.42	42.96±0.44	5.96±0.26	0.324±0.009	8.22±0.56
×	✓	✓	38.34±0.43	52.58±0.46	43.08±0.47	6.18±0.31	0.330±0.010	8.67±0.62
✓	✓	✓	**39.08±0.31**	**53.72±0.36**	**43.52±0.35**	**5.78±0.24**	**0.316±0.009**	**7.82±0.46**

TABLE VI
TARGETED ABLATION OF $\mathcal{L}_{\mathbf{down}}$, WITH ALL OTHER CYTOSET COMPONENTS FIXED. VALUES ARE MEAN ± STD OVER THREE SEEDS. BETHESDA METRICS USE CLASS-AGNOSTIC MATCHES AT IoU≥0.5; EXCEPT MOE (RANK UNITS), ALL VALUES ARE PERCENTAGES.

Setting	mAP↑	AP@0.5↑	AP@0.75↑	SCR↓	MOE↓	HR-Recall↑	HR-DG↓
CytoSet w/o $\mathcal{L}_{\text{down}}$	39.01±0.34	53.60±0.39	43.44±0.39	5.91±0.27	0.320±0.009	89.67±0.67	8.24±0.51
CytoSet	**39.08±0.31**	**53.72±0.36**	**43.52±0.35**	**5.78±0.24**	**0.316±0.009**	**89.86±0.64**	**7.82±0.46**

F. Ablation Study

Table V reports component-wise ablations of DCSR, SGQS, and OCAQ on RT-DETR.

Adding DCSR increased mAP from 37.16 ± 0.36 to 38.08 ± 0.44. SGQS alone primarily improved localization at AP@0.75 and provided no standalone gain in grade-consistency metrics. OCAQ produced the largest standalone changes in Bethesda consistency, reducing SCR from 7.01 ± 0.29 to 6.42 ± 0.30, MOE from 0.360 ± 0.008 to 0.339 ± 0.010, and HR-DG from 10.37 ± 0.72 to 8.94 ± 0.64. The full model achieved the best values across all reported metrics. Adding OCAQ to DCSR+SGQS reduced SCR from 6.86% to 5.78%, MOE from 0.351 to 0.316, and HR-DG from 10.02% to 7.82%, while increasing mAP from 38.88% to 39.08%. This factorial comparison identifies OCAQ as the component associated with the main ordinal-consistency gains in the conditioned-query setting.

The module-level ablation evaluates OCAQ as a unified formulation. Because $\mathcal{L}_{\text{down}}$ has a distinct one-way target, we additionally isolate it while keeping the remaining OCAQ formulation fixed (Table VI).

Removing $\mathcal{L}_{\text{down}}$ left mAP nearly unchanged (39.08 to 39.01) and increased HR-DG from 7.82% to 8.24%; SCR, MOE, and HR-Recall changed only slightly. Thus, $\mathcal{L}_{\text{down}}$ provides a modest direction-specific reduction in high-to-low errors.

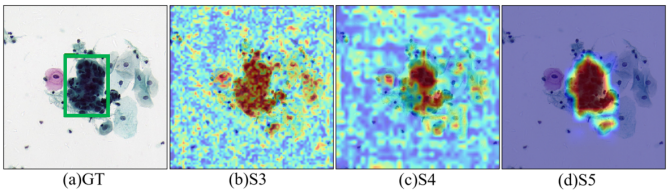


Fig. 5. Feature-level DCSR responses for a TCT patch. (a) Input patch with the annotated target outlined in green. (b)–(d) Response maps from $\boldsymbol{S3}$, $\boldsymbol{S4}$, and $\boldsymbol{S5}$, respectively; warmer colors indicate stronger responses.

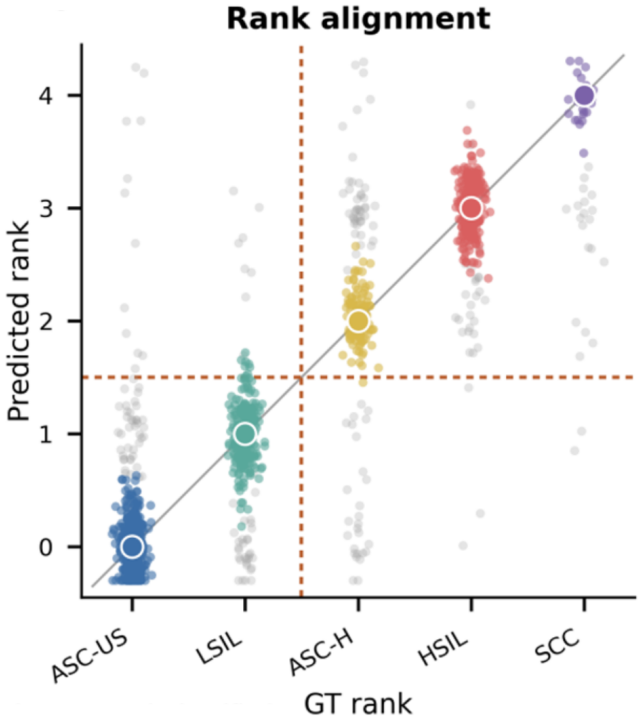


Fig. 6. Center-space rank correspondence of final-layer matched queries. Each query takes the Bethesda rank of its most cosine-similar category center. The axes show ground-truth and center-assigned ranks; colored points indicate agreement, gray points mark errors, and dashed lines mark the low-/high-risk boundary.

Feature-level DCSR responses. Figure 5 compares DCSR response maps across feature levels. In this example, the $S3$ and $S4$ responses extend beyond the annotated target, whereas the $S5$ response is more concentrated around it. This qualitative observation motivates applying DCSR to projected $S5$ while leaving $S3$ and $S4$ unchanged before cross-scale fusion.

G. Mechanism Analysis

1) Center-Space Rank Correspondence: We examine the rank structure of matched queries directly in the OCAQ center space. For each final-layer matched query, we identify the category center with the highest cosine similarity and assign its corresponding Bethesda rank. Figure 6 compares the ground-truth rank with this center-similarity-based rank.

The correspondence is concentrated near the diagonal, with mismatches occurring predominantly between adjacent grades and relatively few long-range cross-grade deviations. Because OCAQ directly optimizes this representation, the visualization is interpreted as a mechanistic sanity check rather than an independent performance measure.

H. Cross-Cohort and External Evaluation

Table VII separates three settings: internal cross-scanner transfer between 0506 and 0818, in-domain replication on the external Tianchi and CRIC datasets, and external transfer from TCT-MS to Tianchi. We report standard detection metrics and localization-controlled Bethesda consistency metrics in each setting.

TABLE VII
CROSS-COHORT TRANSFER AND EXTERNAL DATASET EVALUATION. MATCHING SOURCE AND TARGET DENOTE IN-DOMAIN EVALUATION; DIFFERING DATASETS DENOTE SCANNER OR DATASET TRANSFER. ALL METRICS EXCEPT MOE ARE PERCENTAGES; MOE IS IN ORDINAL-RANK UNITS.

Source	Target	Model	mAP↑	AP@0.5↑	AP@0.75↑	SCR↓	MOE↓	HR-DG↓
0506	0506	RT-DETR	34.0±0.39	43.9±0.44	38.0±0.48	12.36±0.31	0.59±0.010	17.64±0.76
		CytoSet	**35.3±0.34**	**45.8±0.38**	**39.2±0.40**	**11.82±0.26**	**0.56±0.009**	**16.92±0.55**
0506	0818	RT-DETR	21.2±0.54	31.6±0.60	23.5±0.65	24.43±0.44	1.15±0.017	22.86±0.88
		CytoSet	**22.1±0.47**	**32.7±0.52**	**24.4±0.57**	**23.62±0.38**	**1.11±0.015**	**21.74±0.71**
0818	0506	RT-DETR	22.0±0.51	30.8±0.57	24.7±0.61	17.24±0.41	1.09±0.016	20.36±0.84
		CytoSet	**23.1±0.44**	**32.0±0.49**	**25.8±0.54**	**16.41±0.35**	**1.03±0.014**	**19.28±0.68**
0818	0818	RT-DETR	27.8±0.41	39.6±0.46	29.9±0.49	11.96±0.30	0.58±0.010	17.08±0.72
		CytoSet	**28.7±0.35**	**40.8±0.40**	**30.9±0.42**	**11.42±0.25**	**0.56±0.009**	**16.52±0.54**
Tianchi	Tianchi	RT-DETR	28.3±0.47	41.0±0.53	27.0±0.57	12.64±0.36	0.64±0.012	10.52±0.79
		CytoSet	**29.9±0.40**	**43.3±0.46**	**28.5±0.49**	**10.86±0.29**	**0.56±0.010**	**8.72±0.60**
TCT-MS	Tianchi	RT-DETR	15.9±0.60	28.5±0.67	15.1±0.68	27.42±0.50	1.31±0.020	14.08±0.88
		CytoSet	**17.0±0.52**	**30.3±0.59**	**16.3±0.61**	**23.78±0.43**	**1.18±0.017**	**12.20±0.76**
CRIC	CRIC	RT-DETR	27.2±0.50	48.8±0.56	23.1±0.60	11.26±0.39	0.58±0.013	11.10±0.82
		CytoSet	**28.6±0.42**	**50.7±0.48**	**24.2±0.51**	**9.64±0.31**	**0.50±0.011**	**9.35±0.64**

CytoSet outperformed RT-DETR across all evaluated settings, with higher AP and lower SCR, MOE, and HR-DG in both internal and external evaluations.

In the TCT-MS → Tianchi transfer setting, CytoSet increased mAP from 15.9 to 17.0 and reduced SCR from 27.42% to 23.78%, MOE from 1.31 to 1.18, and HR-DG from 14.08% to 12.20%.

The transfer settings were substantially harder than the in-domain settings for both models. This gap coincided with acquisition differences and case-mix shift. As shown in Table II, the internal subsets and external datasets differed markedly in foreground category priors, with the TCT-MS → Tianchi setting reflecting a clear external domain gap.

I. WSI-Level Bethesda Evidence Retention

We conduct detector-derived evidence-retention analysis on 414 internal positive WSIs independent of detector training and 500 Tianchi positive WSIs using sliding windows with 20% overlap. RT-DETR and CytoSet use the main-experiment seed with the highest validation mAP, selected without WSI labels. All detections returned by native inference and post-processing are included in highest-grade aggregation, and the highest grade among all window-level detections defines the WSI prediction.

UNI [52] serves as a patch-level representation baseline. Each training patch is labeled by its highest annotated squamous Bethesda grade; the pretrained encoder is frozen and a linear five-class head is trained on these labels. Its highest predicted patch grade defines the WSI prediction. In both cohorts, the reference is the highest annotated squamous grade.

Table VIII reports WSI-level high-risk downgrading, severe cross-grade error, five-class accuracy, and risk-group accuracy. Compared with RT-DETR, CytoSet reduces WSI-HR-DG from 20.3% to 14.5% internally and from 28.5% to 22.8% on Tianchi. It also lowers WSI-SCR from 9.2% to 7.0% internally and from 20.2% to 19.2% on Tianchi. Five-class accuracy increases from 76.3% to 78.5% and from 56.8% to 61.0%, respectively; risk-group accuracy rises from 88.6% to 90.8% and from 75.2% to 76.2%.

TABLE VIII
WSI-LEVEL BETHESDA EVIDENCE RETENTION ON 414 INTERNAL AND 500 TIANCHI POSITIVE WSIS. VALUES ARE PERCENTAGES, WITH COUNTS IN PARENTHESES.

Cohort	Method	WSI-HR-DG↓	WSI-SCR↓	Accuracy↑	Risk-Group Acc.↑
Internal WSI	UNI	23.9 (33/138)	10.9 (45/414)	73.2 (303/414)	86.5 (358/414)
	RT-DETR	20.3 (28/138)	9.2 (38/414)	76.3 (316/414)	88.6 (367/414)
	CytoSet	**14.5 (20/138)**	**7.0 (29/414)**	**78.5 (325/414)**	**90.8 (376/414)**
Tianchi	UNI	29.9 (84/281)	22.4 (112/500)	60.0 (300/500)	72.6 (363/500)
	RT-DETR	28.5 (80/281)	20.2 (101/500)	56.8 (284/500)	75.2 (376/500)
	CytoSet	**22.8 (64/281)**	**19.2 (96/500)**	**61.0 (305/500)**	**76.2 (381/500)**

V. CONCLUSION

This paper formulates fine-grained TCT squamous cell detection as Bethesda-order-aware set prediction. CytoSet conditions the decoder query set through semantic refinement and spatial selection before applying ordered center alignment and quality coupling. It improves standard detection performance while reducing severe cross-grade errors, mean ordinal error, and high-risk downgrading after localization. The reporting ranks are operational surrogates for cell-level modeling rather than patient-level risks, which also depend on HPV status, screening history, and other clinical factors unavailable in this study. The results support Bethesda-order-aware query learning and indicate improved retention of detector-derived high-grade cellular evidence in the exploratory WSI analysis.

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